Replication Plan

Electronic Specific Heat Capacities and Entropies from Density Matrix Quantum Monte Carlo Using Gaussian Process Regression

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Auto-generated Replication Blueprint

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Contents

Paper Overview

This paper uses density matrix quantum Monte Carlo (DMQMC) and its initiator variant (PIP-DMQMC) to compute finite-temperature properties (internal energy, specific heat capacity, entropy) of small molecular and model systems (e.g., uniform electron gas, small atoms/molecules). Because QMC energies at finite temperature are noisy, the authors apply Gaussian process (GP) regression to smooth the energy-vs-temperature data and then analytically differentiate the GP to obtain the specific heat $C_v(T) = dE/dT$ and integrate to obtain the entropy. They benchmark against exact full configuration interaction (FCI) results.

Detailed Workflow

Phase 1: Exact Diagonalization Benchmarks

1.1 Implement or use FCI for small systems (few electrons in small basis sets).

1.2 Compute exact $E(\beta)$ (energy vs. inverse temperature) via the thermal density matrix:

$$E(\beta) = \frac{\mathrm{Tr}[He^{-\beta H}]}{\mathrm{Tr}[e^{-\beta H}]}$$

1.3 Compute exact $C_v(\beta)$ **and** $S(\beta)$ from the partition function.

Phase 2: DMQMC / PIP-DMQMC Simulations

2.1 Set up the Hamiltonian in second quantization (molecular integrals or UEG parameters).

2.2 Run DMQMC.

- Stochastic propagation of the density matrix in imaginary time.
- Sample at multiple inverse temperatures β values.
- Record $E(\beta)$ with statistical error bars.

2.3 Run PIP-DMQMC (interaction-picture variant) for improved efficiency.

Phase 3: Gaussian Process Regression

3.1 Fit GP to $E(\beta)$ **data.**

- Input: $\{(\beta_i, E_i, \sigma_i)\}$ from DMQMC.
- Kernel: squared-exponential (RBF) or Matérn.
- Heteroscedastic noise model using the QMC error bars.

3.2 Optimize GP hyperparameters (length scale, signal variance) via marginal likelihood.

3.3 Compute derivatives analytically.

- $C_v(\beta) = -\beta^2 \frac{dE}{d\beta}$; the GP derivative is analytic.
- Entropy via thermodynamic integration.

3.4 Propagate uncertainties through the GP to get error bands on C_v and S.

Phase 4: Validation

4.1 Compare GP-derived $C_v(\beta)$ and $S(\beta)$ against exact FCI results.

4.2 Reproduce convergence of DMQMC results with walker number.

4.3 Reproduce all figures: $E(\beta)$, $C_v(\beta)$, $S(\beta)$ with error bands.

Datasets

Dataset / Source	**Type**	**Location / Notes**
Molecular integrals	Generated	From PySCF or similar quantum chemistry code
UEG parameters	Specified	Electron number, r_s, basis set size from paper
DMQMC energy data	Generated	Output of QMC simulations

Models, Tools, and Software

Tool / Library	**Version**	**Purpose / URL**
HANDE	latest	QMC code (DMQMC, FCIQMC) `https://github.com/hande-qmc/hande`
PySCF	≥ 2.0	Molecular integrals, FCI benchmarks `https://pyscf.org/`
scikit-learn	≥ 1.0	Gaussian process regression `https://scikit-learn.org/`
GPy	latest	Alternative GP library `https://sheffieldml.github.io/GPy/`
NumPy / SciPy	≥ 1.21	Numerical operations
Matplotlib	≥ 3.4	Plotting
pyblock	latest	QMC error analysis (blocking/reblocking) `https://github.com/jsspencer/pyblock`

Hardware Requirements

- DMQMC: HPC cluster for large walker populations; small systems feasible on workstation.
- GP regression: any laptop.